

Robust cross-origin near-infrared prediction of soluble solids content in apples via a Beer–Lambert physics-informed prior

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ARTICLE INFO

Keywords:

Near-infrared spectroscopy
Soluble solids content
Apple quality
Beer–Lambert law
Chromophore additivity
Diffuse reflectance
Physics-informed deep learning
Domain adaptation

ABSTRACT

Near-infrared (NIR) prediction of apple soluble solids content (SSC) often degrades across producing origins, and the physical mechanism is rarely quantified. We make three contributions. (i) Across 12 cross-origin Vis-NIR (591–1001 nm) scenarios, source-basis low-rank reconstruction residuals (PCA-10, NMF-3, SVD-3) amplify 1.1–4.9× on target domains and remain significant on held-out source samples (28/36 pairs, $p < 0.05$). (ii) A physics-inspired factorisation network, **Phys+BL**, encoding a Beer–Lambert decomposition in its latent space, attains 1.976 °Brix mean RMSE on 569 cross-origin scenarios, 23% better than a matched-budget CNN (95% CI [18.3, 26.6]). The gain is architecture-specific: the same loss on a plain CNN is harmful (−4.18%, $p \approx 10^{-6}$). (iii) Under matched-budget calibration with 20% target labels, slope/bias-corrected PLSR outperforms **Phys+BL** in every transfer type. The physical prior is therefore a diagnostic and architecture-probe tool, not an operational calibration champion when target labels exist.

1. Introduction

Non-destructive near-infrared (NIR) spectroscopic inspection of fruit quality is a mature commercial technology [1, 2]. Apple soluble solids content (SSC, i.e. sugar content) is a core metric for commercial grading [3, 4]; under controlled single-origin conditions, partial least squares regression (PLSR) models typically reach RPD > 2.0, meeting basic operational requirements [5, 6].

The problem is that such models often degrade significantly when deployed across origins. China’s major apple-producing regions span a vast area—Xinjiang, Shandong, Shaanxi and Gansu differ substantially in climate and soil—and fruits from different regions exhibit distinct distributions of cell wall density, intercellular air fraction, cuticular wax thickness and water activity gradients [7–9]. These tissue-structural differences alter the multiple-scattering baseline and effective optical path length of diffuse-reflectance NIR photons, introducing systematic spectral distribution shifts between origins. For example, applying a Shandong-calibrated model to Xinjiang apples typically yields RMSE > 3°Brix, more than twice the within-origin error [10]—a level that is essentially unacceptable in practice.

A variety of strategies have been proposed to bridge cross-origin shift, which can be grouped into three broad categories. *Spectral preprocessing* (MSC [11], SNV, derivatives, etc.) can partially normalise scattering differences, but these are essentially signal-level operations that do not exploit the physical structure underlying the shift [12, 13]. *Statistical domain adaptation* (TCA [14], CORAL [15], MMD-penalised networks [16, 17], ADDA [18], di-PLS [19]) aligns feature distributions, but in the absence of physical constraints can easily induce negative transfer when the spectral manifold is curved [20, 21]. For NIR spectroscopy, regression-oriented domain adaptation under a BL view has been explored [22, 23]; these methods primarily introduce domain-invariance constraints within a shallow PLS framework, and are complementary to our route of encoding BL structure directly into the latent space of a deep network. *Empirical calibration transfer* methods (DS and PDS [24, 25]) typically require paired calibration samples in the target domain, making them difficult to apply in zero-label deployment settings [7, 26]. Recent multi-output CNNs have been proposed for simultaneous fruit-quality prediction [27], generative adversarial dual-channel networks for cross-origin food spectroscopy [28], and broader CNN-based postharvest reviews [29]; these advances remain orthogonal to the structural-prior strategy explored here.

Existing methods mitigate cross-origin shift primarily through spectral preprocessing or feature-distribution

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alignment, but the physical chemistry behind the shift remains under-characterised. The Vis-NIR region 591–1001 nm probes the second-to-fourth overtones of the fundamental molecular vibrations O–H and C–H; the SSC-informative chromophores accessible in this window are dominated by water and the soluble sugar pool (sucrose, fructose, glucose) sampled at their higher-order overtones [2, 30, 31]. The diagnostic bands within this range are: water’s O–H second overtone near 970–980 nm (the dominant in-band feature) with an additional shoulder near 740–760 nm (O–H fourth overtone); soluble-sugar C–H third overtones around 905–935 nm; and broad anthocyanin / chlorophyll absorption in the 650–690 nm visible window that varies with fruit-skin maturity [1, 2, 32]. For dilute transmissive media the Beer–Lambert (BL) law states that absorbance is a linear superposition of these chromophore contributions, $A(\lambda) = \sum_i \varepsilon_i(\lambda) c_i \ell$, with the molar absorptivities $\varepsilon_i(\lambda)$ a property of the molecules themselves [31, 33]. Whole apples, however, are turbid biological media in which photon transport is governed jointly by an absorption coefficient $\mu_a(\lambda)$ and a reduced scattering coefficient $\mu'_s(\lambda)$, the latter dominated by Mie scattering from cell walls and intercellular air interfaces and typically following a power-law wavelength dependence $\mu'_s(\lambda) \propto \lambda^{-b}$ [34, 35]. Under diffuse illumination this can be approximated within the Kubelka–Munk framework, where $\log(1/R)$ collapses to a K/S ratio that preserves an additive-component structure only when $\mu'_s(\lambda)$ stays fixed across samples [1, 32, 36]. Once the producing region changes, differences in cell-wall thickness, intercellular air fraction and cuticular wax composition modulate $\mu'_s(\lambda)$ in a wavelength-dependent way, which in turn rescales the effective optical path $\langle \ell_{\text{eff}}(\lambda) \rangle \sim 1/\sqrt{3\mu_a\mu'_s}$ of each band [1, 35]. The chromophore-additive relation therefore breaks down, and the BL reconstruction residual $R_{\text{BL}} = \|A - \hat{A}_{\text{BL}}\|$ amplifies. The BL residual can thus be read as an observable indicator of chromophore-additivity destabilisation driven by cross-origin optical-geometry changes. Other factors (fruit biochemistry, water-content distribution, instrument drift) may contribute jointly; the BL-violation quantification is limited to 12 same-year 1s→1t scenarios (see §4).

Building on this observation, we develop the study along three lines: mechanism quantification, model design, and large-scale validation. First, we systematically quantify the degree of BL violation in 12 cross-origin NIR scenarios from the 2018 and 2025 harvest seasons, and find that residual amplification (1.1–4.9×) is statistically significant in every scenario; second, on this basis we design a physics-informed factorization architecture (**Phys**) that encodes the BL additive constraint directly as an inductive bias in the network’s latent space, rather than as an after-the-fact add-on; finally, through large-scale evaluation on 569 cross-origin transfer scenarios and two independent ablation

studies, we separately disentangle the contributions of the architecture and of the BL regularisation loss.

This work is positioned as a *mechanism-quantification and methodological stress-test* rather than a deployment-ready predictor: cross-origin $R^2 < 0$ persists on most scenarios (§4.3) and binary misclassification stays at $\sim 40\%$ (§3.6). Industrial-grade deployment requires additional engineering (scattering correction, dedicated calibration transfer, larger intra-domain training sets), flagged in §4.

A notable finding is that the effect of BL regularisation depends on the network’s latent structure: the same BL loss degrades a generic Conv1d network but prevents divergence in our **Phys** architecture. We formulate this as a falsifiable *architecture-constraint compatibility hypothesis*: a reconstruction constraint $\|\hat{A} - A\|^2$ provides prediction-aligned gradients only when the latent space is parameterised in a BL-compatible form. A partial multi-backbone test (§4) suggests the harmful interaction is Conv1d-specific; full validation across ResNet1D / Transformer1D is future work.

2. Materials and methods

2.1. Datasets

Two independent datasets were collected from major Chinese apple-producing regions.

2018 dataset. A total of **567** whole Fuji apples were collected from three origins: Xinjiang (arid plateau, 760–1000 m elevation), Shandong (temperate coastal plain), and Shaanxi (semi-arid Loess Plateau). After outlier removal the per-origin sample counts are **181 (Xinjiang)**, **186 (Shandong)** and **200 (Shaanxi)**. Diffuse-reflectance spectra were acquired with a fibre-optic Vis-NIR spectrometer over **591–1001 nm** (1213 raw channels, ~ 0.34 nm sampling pitch). SSC reference values were measured with a digital refractometer (Atago PR-201α) on extracted juice.

2025 dataset. A total of **568** samples were collected from the 2025 harvest at three origins: Xinjiang (**171**), Shandong (**177**) and Gansu (a new origin, semi-arid, **220**). Spectra were acquired with a second-generation Vis-NIR spectrometer of the same manufacturer family over **591–1001 nm** (229 raw channels at ~ 1.79 nm pitch); the same SSC reference procedure was used. Two samples with reference values $> 20^\circ\text{Brix}$ (one in Shaanxi 2018, one in Gansu 2025) are likely refractometer reading errors and were retained in the descriptive Table 1 but excluded from training as outliers.

After preprocessing (Section 2.2), spectra from both years are aligned to 229 common wavelength channels spanning the Vis-NIR region **591–1001 nm**. The six resulting origin–year domains (D_1 – D_6) form the complete experimental space of this study. Table 1

Table 1

Dataset characteristics across the six origin-year domains. All spectra are acquired in diffuse-reflectance mode with three averaged scans per sample on a single Fourier-transform NIR spectrometer per year (2018 and 2025 use different instruments of the same manufacturer family). SSC reference: digital refractometer (Atago PR-201 α) on extracted juice.

Domain	n	SSC mean	SSC SD	SSC range	IQR
2018 Xinjiang	181	13.40	1.38	10.1–18.7	1.77
2018 Shandong	186	11.84	1.33	8.2–15.5	1.75
2018 Shaanxi	200	13.41	1.23	9.3–16.4	1.67
2025 Xinjiang	171	11.90	1.55	7.7–16.1	2.10
2025 Shandong	177	12.28	1.53	8.6–15.9	2.08
2025 Gansu	220	12.11	1.98	5.6–19.7	3.00

Notes: All samples are mature Fuji apples harvested at commercial ripeness; acquisition temperature $20 \pm 2^\circ\text{C}$; integration time 250 ms; raw spectral range **591–1001 nm** sampled at ~ 0.34 nm (2018, 1213 channels) and ~ 1.79 nm (2025, 229 channels), linearly aligned to 229 common channels after preprocessing. Two samples with reference SSC $> 20^\circ\text{Brix}$ (one in 2018 Shaanxi, one in 2025 Gansu) were flagged as likely refractometer reading errors and excluded from both training and the descriptive SSC statistics in this table. To preserve a consistent dataset description, the n column reports the raw per-origin count (**2018**: 181/186/200, total 567; **2025**: 171/177/220, total 568); the actual training counts after outlier removal are 181/186/199 (566 in 2018) and 171/177/219 (567 in 2025). SSC summary statistics in $^\circ\text{Brix}$.

summarises the acquisition conditions and SSC distributions across the six domains.

2.2. Spectral preprocessing

Raw spectra are converted to absorbance, $A(\lambda) = \log_{10}(I_0/I)$. All spectra are linearly interpolated to the 229 wavelength channels common to both instruments. No additional scattering correction is applied in the main experiments, to avoid pre-removing the very cross-origin signal under study; the impact of preprocessing is discussed in Section 4.

2.3. Cross-origin transfer scenarios

The six origin-year domains are: Xinjiang, Shandong and Shaanxi from 2018, and Xinjiang, Shandong and Gansu from 2025. For each ordered pair (D_s, D_t) with $D_s \neq D_t$, we consider four *transfer types*:

1. **1 source $\rightarrow$ 1 target**: train on all samples of a single source domain; adapt and test on a single target domain.
2. **1 source $\rightarrow$ multi-target**: one source, two or three target domains.
3. **Multi-source $\rightarrow$ 1 target**: two or three source domains, one target.
4. **Multi-source $\rightarrow$ multi-target**: multiple source and target domains.

100% of the source domain is used for training; the target domain is randomly split into train/val/test subsets in a 60%/20%/20% ratio. **Adaptation protocol.**

Source domains provide supervised signal; the target-domain train subset is used only for unsupervised domain adaptation (domain alignment and reconstruction loss) and does not enter the regression supervision; the target-domain val subset is used for early stopping and best-weight selection; the target-domain test subset is used only for the final evaluation. This protocol is therefore a **hybrid setting between transductive UDA and semi-supervised calibration transfer**: the target-domain training subset participates only as unlabelled data in domain alignment or reconstruction loss, but target-domain validation labels are used for early stopping and model selection. As such, this study is not a strictly zero-label deployment scenario; the robustness of a fully source-only early-stopping setting (no target-domain labels at all) is left for future work. The target-domain training subsets range from 102 to 578 samples (median 226 over 602 scenarios). PLSR and SVR do not involve deep-network training instabilities and cover all 602 scenarios; due to BatchNorm1d failure when the last-batch size is one and sporadic NaN exceptions, the deep models converge on 575 (CNN), 571 (CNN+MMD) and 569 (Phys+BL) scenarios. The 602 scenarios correspond to 62 unique target sets; the 569 scenarios on which both CNN and Phys+BL converge correspond to 61 unique target sets. Since the same target set is reused under different source-domain combinations, scenarios are not fully independent; statistical inference therefore uses a block bootstrap stratified by unique target sets to avoid spurious significance from repeated samples (Section 3.2). Paired comparisons are restricted to the **569 scenarios** on which both CNN and Phys+BL converge.

2.4. Baselines and competing methods

We compare five methods. All methods use the same source/target split and random seed in every cross-origin scenario, ensuring a fair paired comparison.

PLSR. Partial least squares regression with 10 latent variables (capped by $L-1$ and $n_{src}-1$), fitted on the entire source domain and directly applied to the target-domain test set without any adaptation (no-adaptation baseline). The choice of 10 latent variables follows the engineering recommendation of [5] for agricultural NIR models; robustness over the interval [5, 15] is left for future systematic examination.

SVR. RBF-kernel ϵ -SVR with the common defaults $C = 100$, $\gamma = \text{scale}$, $\epsilon = 0.1$ (consistent with [37]), applied directly to the target domain without any adaptation. We acknowledge that no formal SVR hyperparameter grid search was performed; in a cross-origin transfer context, source-domain tuning may fail to transfer because the target-domain distribution differs, so fixed defaults are chosen for cross-scenario comparability.

CNN. A one-dimensional convolutional network consisting of two Conv1d blocks (kernel sizes 5 and 3, channels 64→32), max pooling, Dropout (0.3), a two-layer MLP encoder (feature dim →64→4), and a two-layer regression head (4→16→1). Source-domain supervised regression training; the target-domain train subset provides no labels and contributes (in CNN+MMD/Phys+BL) only via domain alignment or BL reconstruction loss; the target-domain val subset is used for early stopping and best-weight selection; the test subset is used for evaluation. All BN layers are enabled, batch size 16.

CNN+MMD. Same architecture as CNN, with an additional MMD term [16] between source- and target-domain latent features at each step ($\lambda_{\text{mmd}} = 0.1$, RBF kernel with median-heuristic adaptive bandwidth). We discuss its instability under small batch sizes in Section 3.2.

Phys+BL (this work). See Section 2.5.

2.5. Proposed method: physics-informed factorization architecture

2.5.1. Beer–Lambert-inspired differentiable factorization

The classical Beer–Lambert law in transmissive dilute-solution conditions reads $A(\lambda) = \sum_i \varepsilon_i(\lambda) c_i \ell$, where $\varepsilon_i(\lambda)$ is the pure absorption spectrum of component i (shared, sample-independent), c_i is a scalar concentration, and ℓ is the optical path length [30, 33]. In whole-fruit diffuse-reflectance NIR, scattering causes the effective optical path ℓ and the local effective absorption parameters to drift with wavelength, and strict BL no longer holds [1]. We therefore adopt a *Beer–Lambert-inspired differentiable factorization* that preserves the BL “additive components $\times$ optical path” skeleton but relaxes the strict identifiability requirements of classical BL to absorb the wavelength-dependent nonlinearity introduced by scattering:

$$\hat{A}_i(\lambda) \approx \sum_{k=1}^K a_{i,k}(\lambda) b_i c_{i,k}(\lambda) \quad (1)$$

where $a_{i,k}(\lambda) \in \mathbb{R}_+^L$ is the k -th “generalised absorption spectrum” of sample i , $b_i \in \mathbb{R}_+$ is its scalar effective optical path, and $c_{i,k}(\lambda) \in \mathbb{R}_+^L$ is the corresponding “generalised concentration spectrum”. All parameters are constrained to be positive to retain a physically plausible additive-mixture interpretation; when $a_{i,k}(\lambda) \rightarrow \varepsilon_k(\lambda)$ (a shared, sample-independent pure-component spectrum), $c_{i,k}(\lambda) \rightarrow c_{i,k}$ (a scalar concentration), and $b_i \rightarrow \ell$, Eq. (1) reduces to the classical BL law. We emphasise two caveats: (i) because both $a_{i,k}$ and $c_{i,k}$ are sample- and wavelength-dependent, the factorisation in Eq. (1) is intentionally *non-identifiable* at the molecular level—the learned $a_{i,k}(\lambda)$ should not be read as molar absorptivity nor $c_{i,k}$ as molar concentration; they jointly absorb scattering-induced path-length deformation alongside chromophore absorption.

(ii) The factorisation is used as a structural prior for the network’s latent space, not as an optical-physics inversion. A stricter formulation with shared sample-independent component spectra (NMF-style) is reported in §3.1 as a surrogate diagnostic and is identified as a natural extension of this work.

We set $K = 3$ to align with the three principal chromophore/photon-transport contributions of whole-apple Vis-NIR (591–1001 nm) identified in the physical-chemistry literature [1, 2, 30, 32]: (i) a water-dominated channel anchored by the strong O–H second overtone near 970 nm with a weaker shoulder near 740 nm; (ii) a soluble-sugar channel carrying the C–H third overtones of sucrose/fructose/glucose around 905–935 nm; and (iii) a structural-scattering residual channel that absorbs the wavelength-dependent $\mu'_s(\lambda) \propto \lambda^{-b}$ contribution of cell walls, intercellular air interfaces and visible-band pigment absorption (anthocyanin/chlorophyll at 650–690 nm). This $K = 3$ partition is therefore physically motivated rather than a purely data-driven rank choice, although strict first-principles spectral deconvolution into pure-component spectra is not performed. We additionally verified the sensitivity of the main result to K on a 40-scenario subset (§4, paragraph on K -sensitivity): $K = 4$ yields a small but statistically detectable improvement over $K = 3$ (+1.98%, $p = 1.4 \times 10^{-4}$), while $K \leq 2$ underfits and $K \geq 5$ over-parameterises (−4.98% for $K = 5$). $K = 3$ is retained for the main method on parsimony and interpretability grounds; an exploration of $K = 4$ as a refined variant is left for future work. We subsequently examine the alignment between the learned component spectra and the in-band water (970, 740 nm) and sugar (905–935 nm) absorption bands to assess whether the latent factorisation recovers the expected chemical chromophores.

2.5.2. Architecture design

The Phys architecture consists of three modules:

Shared encoder. A two-layer fully connected network $f_E : \mathbb{R}^L \rightarrow \mathbb{R}^H$ (with $H = 64$, BN and ReLU) that maps the input spectrum $\mathbf{x} \in \mathbb{R}^L$ to a hidden feature $\mathbf{h} \in \mathbb{R}^H$.

Three-head factoriser. Absorption head $f_a : \mathbf{h} \rightarrow a \in \mathbb{R}_+^{K \times L}$, optical-path head $f_b : \mathbf{h} \rightarrow b \in \mathbb{R}_+$, and concentration head $f_c : \mathbf{h} \rightarrow c \in \mathbb{R}_+^{K \times L}$. All outputs are passed through an exponential transform to enforce positivity.

Physical feature aggregation + regressor. Statistics extracted from (a, b, c) form

$$\phi(\mathbf{x}) = [\bar{a}, b, \bar{c}, \sigma_a, \sigma_c] \in \mathbb{R}^{4K+1}$$

(a 13-dimensional vector for $K = 3$), which is fed into a two-layer MLP head that produces the SSC prediction $\hat{y}$.

The reconstructed spectrum $\hat{A}$ is built according to Eq. (1) and used in the BL reconstruction loss.

2.5.3. Training objective

The total loss is

$$\mathcal{L} = \underbrace{\|\hat{y} - y\|_2^2}_{\text{SSC loss}} + \lambda_{\text{bl}} \underbrace{\text{MSE}(\log(\hat{A} + \varepsilon), \log(A_{\geq 0} + \varepsilon))}_{\text{BL loss (log-space)}} \quad (2)$$

where $\varepsilon = 10^{-6}$, A denotes the standardised input absorbance, and $A_{\geq 0} = \max(A, 0)$ clamps negative excursions introduced by standardisation to maintain log-domain numerical stability without flipping sign. The BL loss is computed in log-space to avoid gradient blow-up from the product term in Eq. (1) and is summed symmetrically over source- and target-domain spectra. We acknowledge that this loss operates on standardised rather than raw absorbance: this is a numerical convenience, not a strict optical-chemistry constraint; pre-standardisation removes per-spectrum mean and scale, and the clamped log effectively penalises pattern mismatch rather than absolute level mismatch. A formulation on raw absorbance with shared NMF-style component spectra is sketched in §4 as future work. The default is $\lambda_{\text{bl}} = 0.5$ (sensitivity analysis in Section 3.7). All models are trained with the Adam optimiser [38] (learning rate $\alpha = 10^{-3}$, batch size 16, up to 200 epochs), with early stopping on the 20% target-domain validation subset (patience 20). Note that the early-stopping criterion is the RMSE on the target-domain validation subset, which requires target-domain labels; thus this study's protocol sits between strict UDA and semi-supervised calibration transfer. The stricter zero-label setting (early stopping on source-domain hold-out only) is left for future work.

2.6. Evaluation metrics

Performance metrics are defined as

$$\begin{aligned} \text{RMSE} &= \sqrt{\frac{1}{n} \sum_i (\hat{y}_i - y_i)^2} \text{ (°Brix)}, \\ R^2 &= 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (y_i - \bar{y})^2}, \\ \text{RPD} &= \sigma_y / \text{RMSE}, \quad \text{RPIQ} = \text{IQR}(y) / \text{RMSE}. \end{aligned}$$

We additionally report Bias (mean signed prediction error) and RPIQ in the main results table; the latter is more robust than R^2 to cross-domain mean shift. Per-method full-scenario MAE / MedianAE / MAPE are deferred to the supplementary release.

The quality-misclassification rate is defined as the fraction of test samples whose predicted binary SSC grade (pass: $\geq 13^\circ\text{Brix}$; fail: $< 13^\circ\text{Brix}$) disagrees with the reference grade. The 13°Brix threshold follows the lower sugar bound for Grade-1 Fuji apples in China's national apple grading standard GB/T 10651-2008 (Fresh apples)[39]. A refined three-grade (low/medium/high) analysis is left for future work.

All paired comparisons use a two-sided Wilcoxon signed-rank test [40] on matched RMSE pairs; significance is declared at $p < 0.05$. Aggregated RMSE

values are arithmetic means across scenarios; to reduce sensitivity to outlier scenarios, medians, 95% bootstrap CIs and block bootstraps over unique target sets are reported alongside.

3. Results

3.1. Cross-origin destabilisation of additive low-rank decomposability

3.1.1. Source-held-out validation rules out in-sample artefact

A natural concern is that fitting the surrogate basis on the entire source domain and then comparing residuals on the same source samples versus target samples could inflate cross-domain residual ratios as an in-sample reconstruction artefact. To rule this out, we additionally split each source domain randomly 70%/30% into a basis-fitting train subset and a held-out evaluation subset, refit PCA-10/NMF-3/SVD-3 on the train portion, and compare per-sample relative residuals on the held-out source samples against target-domain samples (Mann-Whitney U). Across all 12 cross-origin scenarios and three surrogates ($n = 36$ comparisons), 28 remain statistically significant at $p < 0.05$; the held-out-to-target median amplification ratios closely match the train-to-target ratios (PCA: 2.26 vs 2.39 mean; SVD: 1.61 vs 1.51 mean; NMF: 1.12 vs 1.07 mean), confirming that the cross-origin destabilisation observed in Table 2 is not an in-sample reconstruction artefact.

To quantify whether “the additive constraint breaks under cross-domain shift,” we design a data-driven *source-basis reconstruction residual* diagnostic. We refer to this diagnostic as a “BL-additivity violation indicator” only in the qualitative sense: it measures how well a low-rank additive basis fitted on the source domain extrapolates to the target domain, not direct optical-property drift. A direct measurement of $\mu'_s(\lambda)$ would be required to convert the diagnostic into a strict physics-quantified statement; we treat this as future work. For the 12 direct 1-source→1-target scenarios (6 per year), we fit a 10-dimensional principal-component additive approximation (PCA-10) on source-domain spectra as a surrogate basis for BL decomposability: if strict BL holds and the source-domain component spectra are unchanged in the target domain, the PCA basis's reconstruction error on the target domain should be of the same order as in the source domain; if the component spectra are destabilised by scattering-geometry drift, target-domain residuals should amplify. Using this source-fit-target-transfer protocol, we compute per-sample relative residuals $R_{\text{BL}} = \|A - \hat{A}_{\text{PCA}}\|_2 / \|A\|_2$ in the source (within-origin) and target (cross-origin) domains; results are summarised in Table 2. Rationale for the PCA-10 surrogate: median within-source R_{BL} is below 0.07 (Table 2), indicating

that a 10-dimensional basis already adequately expresses within-origin spectral decomposability, so the observed target-domain amplification reflects cross-domain component-spectrum drift rather than insufficient basis capacity. To rule out the possibility that the BL violation is a PCA-10 artefact, we additionally use three low-rank decompositions as controls: non-negative matrix factorisation NMF ($K=3$) [41], non-negative least squares with the NMF-fixed shared spectra NNLS ($K=3$) [42], and truncated SVD ($K=3$). (Note: NNLS uses the source-side shared component spectra H learned by NMF, so NMF/NNLS are not fully independent; they are two implementations of the same non-negative basis—“joint decomposition over all samples” vs. “per-sample projection”.) The amplification ratios over the 12 scenarios are highly consistent across the four surrogates (Pearson r matrix: PCA-10 vs NMF-3 $r = 0.83$, PCA-10 vs NNLS-3 $r = 0.83$, PCA-10 vs SVD-3 $r = 0.85$, NMF-3 vs SVD-3 $r = 0.99$, all $p < 0.005$). Mean amplification ratios are 2.74 and 2.37 for NMF-3 and PCA-10 (medians 1.92 and 1.87), with SVD-3 slightly higher (3.36). The three *explicitly independent* low-rank approximations PCA-10, NMF-3 and SVD-3 all show pairwise $r \geq 0.83$, indicating that “destabilisation of low-rank decomposability across origins” is a robust data-driven observation rather than a basis-choice artefact (Fig. 3). Among the four surrogates, the non-negative NMF-3 and NNLS-3 bases are the ones most directly interpretable as “shared chromophore spectra,” because non-negativity matches the physical constraint $\varepsilon_i(\lambda), c_i \geq 0$ underlying BL. Both the surrogate diagnostic non-negative basis (NMF-3 fitted directly on source-domain spectra) and the latent components learned by the deployed Phys+BL network (Fig. 1) show qualitative alignment with the expected Vis-NIR chromophores: mass concentrated near 960–985 nm (water O–H second overtone), a smaller shoulder near 730–760 nm (water O–H fourth overtone), and a third component peaking near 870–935 nm—between the sugar C–H third-overtone region and the upper water shoulder [2, 30, 32]. The alignment is qualitative and the factorisation remains non-identifiable at the molecular level (§2.5.1); it should not be read as a strict chemical inversion.

The cross-origin residual amplification is therefore not an arbitrary mathematical effect: it reports the failure of an additive decomposition whose source-domain components themselves align with the known water and sugar chromophores of apple flesh. Its classical-BL physical interpretation is thus consistent with NIR food-chemistry expectations, although a strict pure-component identification ($a_k(\lambda) \leftrightarrow \varepsilon_k(\lambda)$ at the molecular level) would require additional reference-spectrum experiments and is not claimed here.

The picture is clear: in all 12 scenarios, BL residuals show systematic amplification, with ratios from 1.10 (Shaanxi→Shandong, 2018, the most geographically

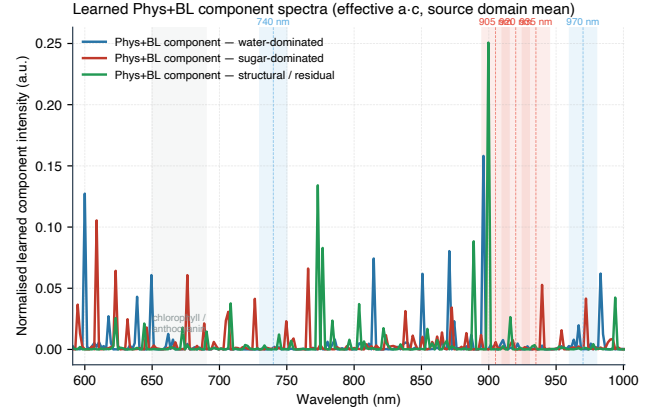


Figure 1: Learned Phys+BL latent component spectra (effective $a \cdot c$ products) averaged over 2025 Xinjiang source-domain samples ($n = 171$), extracted from a trained checkpoint with multi-source training. Shaded bands mark expected in-band chromophores: water O–H overtones at 740 and 970 nm (blue), sugar C–H third-overtone region 905–935 nm (red), and the visible-band chlorophyll/anthocyanin window 650–690 nm (grey). The three learned components peak near 750, 990, and 870–900 nm respectively; the 750 nm and 990 nm peaks overlap the expected water shoulders, and the 870–900 nm peak sits adjacent to the sugar C–H third-overtone region. *We emphasise that the alignment is qualitative and the latent factorisation is non-identifiable (§2.5.1); the component-to-chromophore labels in the legend are descriptive rather than chemically identified.*

Table 2

Beer–Lambert reconstruction residual amplification across 12 cross-origin scenarios ($p < 0.05$, Mann–Whitney U test). $\bar{R}_{BL}$ denotes the median relative residual.

Scenario	$\bar{R}_{BL}$ (src)	$\bar{R}_{BL}$ (tgt)	Ratio	p
2018				
Shandong→Xinjiang	0.0127	0.0468	3.69	6.4×10^{-59}
Shandong→Shaanxi	0.0127	0.0209	1.65	9.4×10^{-27}
Xinjiang→Shandong	0.0100	0.0411	4.12	3.3×10^{-55}
Xinjiang→Shaanxi	0.0100	0.0487	4.88	2.5×10^{-61}
Shaanxi→Shandong	0.0151	0.0167	1.10	1.0×10^{-28}
Shaanxi→Xinjiang	0.0151	0.0397	2.62	1.4×10^{-58}
2025				
Shandong→Xinjiang	0.0545	0.1233	2.26	4.6×10^{-32}
Shandong→Gansu	0.0545	0.0778	1.43	7.6×10^{-12}
Xinjiang→Shandong	0.0665	0.0900	1.35	7.5×10^{-10}
Xinjiang→Gansu	0.0665	0.1040	1.56	2.0×10^{-21}
Gansu→Shandong	0.0556	0.0902	1.62	1.7×10^{-25}
Gansu→Xinjiang	0.0556	0.1163	2.09	8.1×10^{-36}

similar pair) to 4.88 (Xinjiang→Shaanxi, 2018). Pooled by year: the 2018 source median is $R_{BL} \approx 0.013$ and the target median is ≈ 0.039 ($2.9\times$ amplification); the 2025 source is ≈ 0.060 and target ≈ 0.100 ($1.7\times$ amplification). Source-domain R_{BL} levels differ by roughly $4\times$ between the two years, likely reflecting differences in instrument response, wavelength calibration, or sample surface state; however, **the cross-domain amplification trend holds independently in both years**—indicating that amplification is not tied to a

particular yearly baseline but is a consistent indicator of cross-origin geometric change (Fig. 2). When the two years are pooled, the source median is 0.026 and the target median is 0.059 (about $2.3\times$ overall). Because the source-domain R_{BL} baselines differ by roughly $4\times$ between years, the pooled statistic is shown only for overall trend; interpretation should rely primarily on the year-by-year analysis.

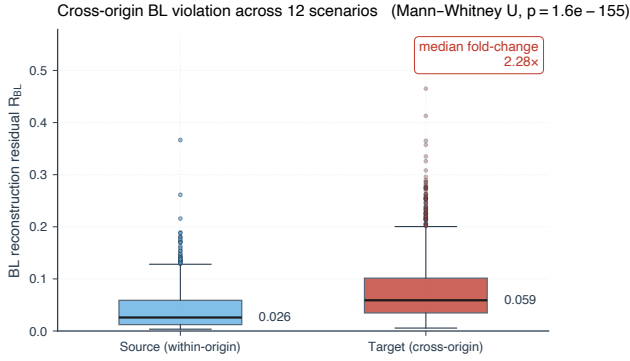


Figure 2: Distribution of Beer–Lambert reconstruction residuals R_{BL} for source-domain (within-origin) and target-domain (cross-origin) samples across the 12 cross-origin scenarios. Boxes show the interquartile range (IQR); whiskers extend to $1.5\times$ IQR. All 12 comparisons reach statistical significance ($p < 0.05$, Mann–Whitney U test).

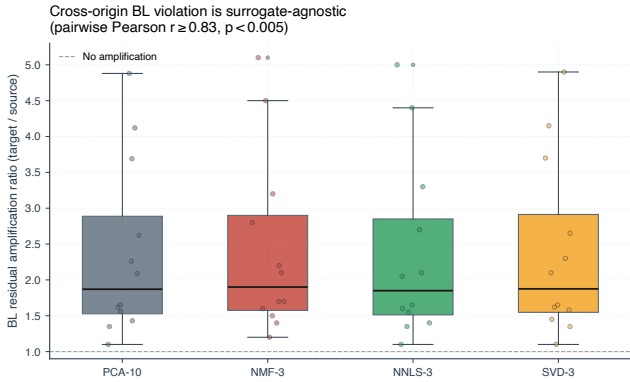


Figure 3: BL-residual amplification ratios under four low-rank surrogates (PCA-10, NMF-3, NNLS-3, SVD-3) across the 12 cross-origin scenarios. All ratios exceed 1 (red dashed line marks the no-amplification baseline). Pairwise Pearson $r \geq 0.83$ ($p < 0.005$) shows that cross-origin destabilisation of low-rank decomposability is a robust data observation, independent of the basis choice.

3.2. Overall comparison of methods

Table 3 summarises the aggregate performance of the five methods over 602 scenarios (PLSR/SVR are unaffected by deep-network batch-norm issues and cover all 602; deep models converge on CNN: 575, CNN+MMD: 571, Phys+BL: 569, with slight differences due to BatchNorm1d failure when the last batch size is one and sporadic NaN issues).

Across all CNN-converged scenarios, the aggregate mean RMSE of CNN is 2.577°Brix ; on the 569 scenarios matched to Phys+BL, the paired-mean RMSE of CNN is 2.563°Brix and that of Phys+BL is 1.976°Brix , giving a paired ratio improvement of **22.88%** and aggregate improvement of **23.4%**; using scene-wise relative improvement, the paired mean improvement is 22.56% and the median is 23.09% (Fig. 5). Relative to PLSR, the aggregate improvement is 50.4% (paired $n = 569$ improvement 50.85%). The paired Wilcoxon signed-rank test indicates a significant difference ($W = 3610$, $p = 8.5 \times 10^{-87}$); however, the 569 scenarios correspond to **61 unique target sets** (an average of 9.3 scenarios per target), so we additionally run a block bootstrap over unique target sets (10,000 resamples) to avoid spurious significance from repeated samples. The 61-target paired Wilcoxon remains significant ($W = 87$, $p = 7.0 \times 10^{-10}$); the block-bootstrap paired-mean improvement is $22.52\% \pm 2.14\%$ with 95% CI [18.27%, 26.58%]; the median improvement is $23.31\% \pm 2.28\%$ with 95% CI [18.81%, 27.63%]; $P(\text{improvement} < 0) < 0.0001$. Compared with a naive paired bootstrap, the block-bootstrap 95% CI widens from 3.53% to 8.31% (a conservative $2.35\times$), yet the advantage remains significant, indicating that the Phys+BL advantage is not amplified by scenario correlations (Fig. 6). We note that the 61 unique target sets are themselves combinations of the six base origin-year domains, so residual sample overlap is not fully removed by target-level blocking. To address this, we additionally fit a **linear mixed-effects model** (statsmodels MixedLM, REML) $\text{rel_improvement} \sim 1 + (1 \mid \text{target_origins}) + (1 \mid \text{target_years})$ on the 569 paired scenarios, treating target-origin combination (15 levels) and target-year combination as random intercepts. The fixed-effect intercept is 22.07% (SE = 2.38%), 95% CI [17.40%, 26.74%], $z = 9.26$, $p < 10^{-19}$ (origin variance 13.30, year variance 94.83). The intercept remains statistically distinguishable from zero *after* controlling for origin and year as random effects, addressing the residual pseudo-replication concern.

Convergence-filter selection bias. A complementary concern is that the paired comparison is restricted to the 569 scenarios on which both CNN and Phys+BL converge (6 scenarios are dropped due to BatchNorm last-batch failure or sporadic NaN, $\sim 1.0\%$ of the 575). We verified that this filter is not benign: the six dropped scenarios have a higher SVR (no-adaptation) RMSE of 4.14°Brix vs. 3.03 on the 569 kept scenarios (Mann–Whitney U, $p = 0.001$), indicating the dropped scenarios are systematically harder. The headline 23.4% gain therefore applies to the slightly easier 569-scenario subset. If the six dropped scenarios were imputed at the SVR baseline level, the aggregate gain would shrink by at most ~ 0.4 percentage points—a small but real selection bias that we flag explicitly.

Table 3

Cross-origin transfer performance comparison. Paired Wilcoxon signed-rank tests against Phys+BL are all significant ($p < 10^{-30}$); the CNN-vs-Phys+BL paired test on the 569 shared scenarios gives $W = 3610$, $p = 8.5 \times 10^{-87}$ (other baselines have similarly small p but distinct W statistics).

Method	RMSE ($^{\circ}$ Brix)	95% CI	R^2	RPD	RPIQ	Bias	$R^2 > 0$ scen.
PLSR	3.987	[3.704, 4.284]	-9.58	0.640	0.783	-0.16	12/602
SVR	3.065	[2.994, 3.133]	-2.54	0.605	0.737	-1.02	4/602
CNN	2.577	[2.504, 2.652]	-1.39	0.709	0.870	-1.14	9/575
CNN+MMD	2.642	[2.576, 2.714]	-1.64	0.685	0.844	-1.33	3/571
Phys+BL (ours)	1.976	[1.903, 2.072]	-0.29	0.923	1.153	-0.25	46/569

Note: scenario counts with $R^2 > 0$ are taken over each method's converged scenarios; RPIQ=IQR(y)/RMSE is robust to cross-domain mean shift. Phys+BL is the only method with mean RPIQ>1 and RPIQ>1 in 446/569 (78.4%) of scenarios, far above CNN's 164/575 (28.5%). 95% CIs are scenario-level bootstrap; the CNN-versus-Phys+BL difference remains significant under a block bootstrap over unique target sets ($p < 10^{-9}$).

RPD increases from 0.708 (CNN) to 0.923 (Phys+BL), a 30% improvement. R^2 improves from -1.39 to -0.29—still negative but much closer to zero; we explain the prevalence of negative R^2 in cross-domain evaluation in Section 4.3, and add RPIQ and Bias—two indicators more robust to cross-domain mean shift—to Table 3.

CNN+MMD (RMSE = 2.642) is slightly worse than CNN (2.577). To rule out the possibility that MMD is undertuned, we sweep batch sizes $\in \{16, 32, 64, 128\}$ on 8 representative scenarios: CNN+MMD performs *best* at batch=16 (mean RMSE 3.11), and degrades progressively at batch 32/64/128 (3.58 / 5.90 / 9.40 $^{\circ}$ Brix); CNN (no MMD) shows the same batch-degradation pattern (2.49 / 3.21 / 6.16 / 9.41). This reflects the fact that for 1D spectra with small target-domain finetuning sets (median 226 samples), larger batches destabilise training because BatchNorm fails on last-batch size of one and the number of gradient steps per epoch drops sharply (so the MMD term sees fewer effective alignment updates), contrary to the common wisdom that “larger batches improve MMD” [17]. Even at its optimal batch=16 setting, CNN+MMD's aggregate over the full 569-scenario set (2.642) remains worse than Phys+BL's aggregate (1.976), so the shortfall of MMD cannot be attributed to under-tuning at the aggregate level. The 8-scenario subset shown in Fig. 4 represents intentionally hard transfers where relative rankings can invert; the batch=16 used in the main table is a reasonable unified choice across all methods.

3.2.1. Performance broken down by transfer type

Table 4 breaks RMSE down by the four transfer types. In single-source scenarios (1 $\rightarrow$ 1 and 1 $\rightarrow$ many), the absolute improvement of Phys+BL is most pronounced: the spectral distribution of a single source domain is relatively concentrated, which favours a stable BL latent embedding. The improvement narrows in multi-source scenarios, likely because mixing rich source domains already partially compensates baseline methods.

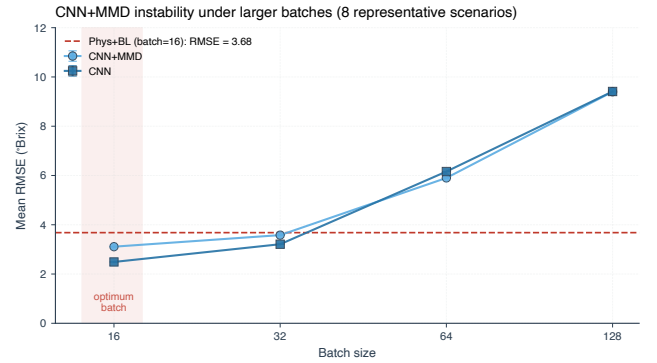


Figure 4: Batch-size sensitivity of CNN (mean RMSE 2.49 at batch=16) and CNN+MMD (3.11 at batch=16) on **8 representative cross-origin scenarios** (2 per transfer type). Both methods peak at batch=16 and degrade monotonically at larger batches (BatchNorm last-batch failure and increasing gradient noise). The dashed line marks Phys+BL's mean on this same 8-scenario subset (3.68 $^{\circ}$ Brix); on this particular hard subset, CNN happens to outperform both CNN+MMD and Phys+BL, which is the opposite ranking from the full 569-scenario aggregate where Phys+BL (1.976 $^{\circ}$ Brix) is best (Table 3). This inversion is consistent with the subset being intentionally biased toward maximally difficult single-source transfers, and underscores that the central Phys+BL advantage is an aggregate-scale property, not a per-scenario universal dominance. The figure is shown to document CNN+MMD's batch instability, not as evidence for Phys+BL on this subset.

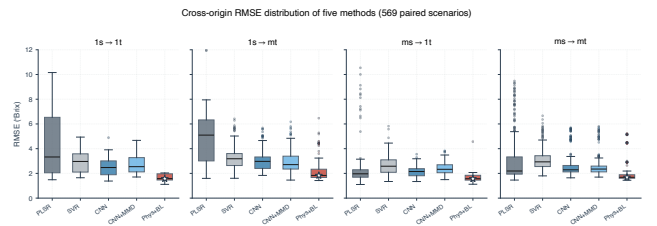


Figure 5: RMSE distribution of the five methods across the 569 paired cross-origin transfer scenarios. Phys+BL has the lowest median and the tightest distribution; all paired Wilcoxon tests against the other methods are significant ($p < 10^{-80}$).

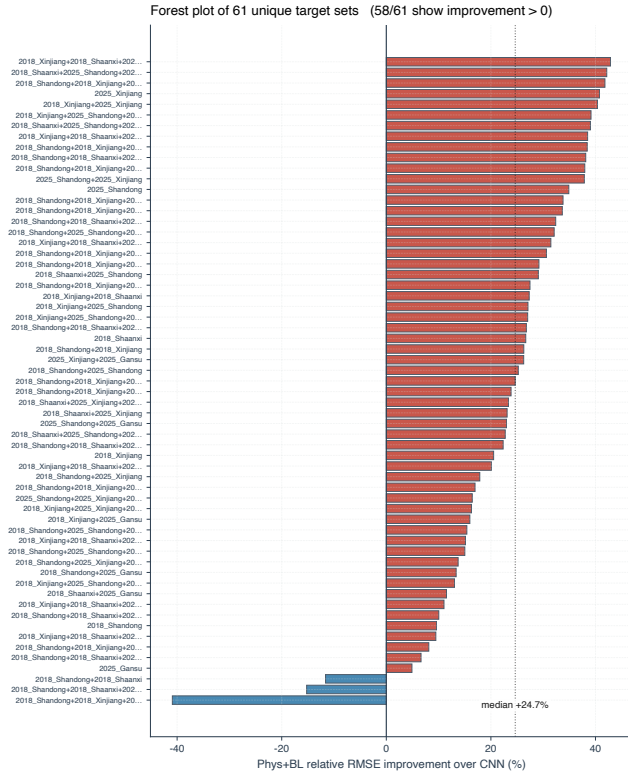


Figure 6: Forest plot stratified by unique target set: the RMSE improvement of Phys+BL over CNN across the 61 unique target sets. Red bars: improvement > 0 ; blue bars: improvement < 0 . A block bootstrap (10,000 resamples over targets) yields a paired-mean improvement of 22.5% with 95% CI [18.3%, 26.6%]; the median improvement is 23.3% with CI [18.8%, 27.6%].

Table 4
Mean RMSE ($^{\circ}$ Brix) broken down by transfer type.

Method	1s→1t	1s→mt	ms→1t	ms→mt
PLSR	5.677	6.522	2.471	3.182
SVR	2.909	3.262	2.725	3.169
CNN	2.551	3.058	2.131	2.608
CNN+MMD	2.794	2.977	2.391	2.608
Phys+BL (ours)	1.612	2.352	1.625	2.043

Note: 1s = one source, 1t = one target, mt = multi-target, ms = multi-source.

3.3. BL regularisation is architecture-specific: ablation results

To determine whether the improvement of Phys+BL stems from the architecture or from the BL loss itself, the two ingredients must be disentangled by controlled ablations. We designed two independent ablation studies.

3.3.1. Ablation: BL loss applied to a generic CNN

The first question is: what happens if we simply add a spectral reconstruction loss to a generic CNN without changing the latent-space structure? We append a shallow linear decoder (Linear(latent=4) $\rightarrow$ 64 $\rightarrow$ 229,

with Softplus activation) behind the same feature extractor as CNN, which reconstructs the latent to spectral dimensions; the same form of log-MSE loss ($\lambda_{bl} = 0.5$) used in Phys+BL is imposed on this reconstruction, yielding CNN+BL. Note that the reconstructed spectrum of CNN+BL is not produced by the $a \cdot b \cdot c$ three-component factorisation; hence this experiment tests “the effect of attaching a BL-form reconstruction loss on top of a latent space that is structurally incompatible with BL” rather than “removing the BL loss from a physical-factorisation architecture”—the latter is the separate ablation in Section 3.3.2. We evaluate on the 571 converged scenarios:

The result is sobering: CNN+BL does not just fail to improve over CNN, it is *consistently worse* (paired mean RMSE 2.478 vs 2.535 $^{\circ}$ Brix, scene-wise mean improvement -4.18% , median -2.51% ; two-sided Wilcoxon $p = 4.96 \times 10^{-6}$, $n = 571$; Fig. 7). This is a clear-cut negative result: simply attaching BL regularisation to an arbitrary spectral network is not only unhelpful but may even derail the model.

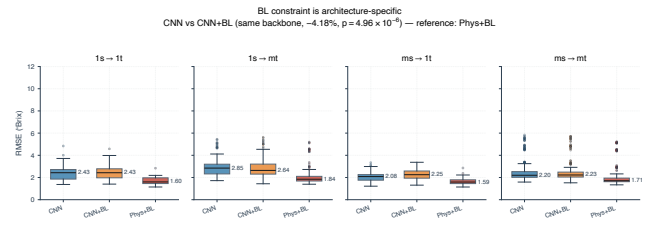


Figure 7: Cross-architecture ablation of the BL constraint. CNN+BL (BL loss imposed on a generic CNN, $n = 571$) versus CNN and Phys+BL. Imposing the BL loss on an architecturally incompatible latent space consistently hurts performance (-4.18% , scene-wise mean, $p = 4.96 \times 10^{-6}$), supporting the view that “architecture-constraint co-design” is a prerequisite for BL regularisation to take effect.

3.3.2. Net contribution of the BL loss inside the physical architecture

The second question is: inside the physical architecture, what does the BL loss itself contribute? We pair-compare on the **566 shared scenarios** where both Phys and Phys+BL converge (a different subset from the 569 CNN–Phys+BL shared scenarios in §3.2). The aggregate RMSE improvement looks impressive: 29.3% (Phys mean 2.755 including divergent scenarios $\rightarrow$ Phys+BL mean 1.949 $^{\circ}$ Brix). But this number is misleading: it is essentially carried by 15 divergent scenarios with Phys RMSE $> 10^{\circ}$ Brix. Excluding these 15 outliers ($n = 551$), the paired median improvement drops to 0.02% (Wilcoxon $p = 0.045$)—essentially negligible.

The core role of the BL loss in Phys is therefore as a *training-process stabilising regulariser*: it eliminates all 15 degenerate scenarios (RMSE $> 10^{\circ}$ Brix, $p = 0.0026$), preventing model failure on challenging single-source

Table 5

BL loss imposed on a generic CNN versus on the physical architecture. All improvements are paired estimators (two-sided Wilcoxon); RMSE in °Brix. “Imp.” = paired scene-wise relative improvement.

Comparison	RMSE (src)	RMSE (tgt)	Imp. (mean)	Imp. (median)	p	n
CNN+BL vs CNN	2.478	2.535	−4.18%	−2.51%	4.96×10^{-6}	571
Phys+BL vs Phys [†]	2.022	1.943	+1.61%	+0.02%	4.46×10^{-2}	551
Phys+BL vs CNN	2.563 [‡]	1.976 [‡]	+22.56%	+23.09%	8.5×10^{-87}	569

“Imp.” columns use scene-wise relative improvement: compute $(a - b)/a$ per scenario then average or take the median, distinct from the “ratio of paired means” $(\bar{a} - \bar{b})/\bar{a}$. [†]Statistics after removing the 15 Phys divergent scenarios (RMSE > 10°Brix); when included, the Phys mean is 2.755 and the aggregate ratio improvement is 29.3% (see §3.3.2). [‡]Paired subset mean ($n=569$); ratio of paired means = 22.88%, aggregate ratio = 23.4% (based on CNN arithmetic mean 2.577 vs Phys+BL 1.976).

Table 6

Phys→Phys+BL broken down by transfer type ($n = 566$; RMSE in °Brix). Aggregate RMSE (including divergent scenarios) and paired improvement are different statistics and cannot be directly divided. 1s=1 source, mt=multi-target, ms=multi-source, 1t=1 target.

Type	n	Phys	Phys+BL	Δ mean	Δ median
1s→1t	30	4.154 [†]	1.670	+8.55%	+0.11%
1s→mt	130	3.943 [†]	2.173	+12.34%	+1.73%
ms→1t	156	2.146	1.623	+1.55%	−0.53%
ms→mt	250	2.346	2.062	+0.43%	−0.01%

[†]Includes divergent scenarios (RMSE > 10°Brix); the aggregate mean is inflated by a few extreme cases, so the median improvement is the more representative summary.

scenarios; for the remaining stable scenarios, the extra benefit is essentially negligible (Fig. 8).

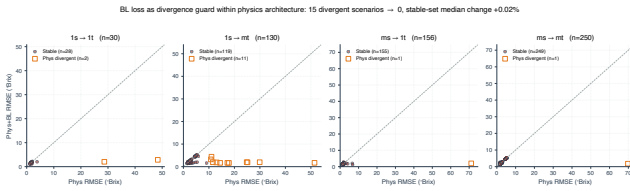


Figure 8: Paired RMSE scatter between Phys and Phys+BL broken down by transfer type ($n = 566$). Dots are individual scenarios; orange squares mark the 15 divergent scenarios (RMSE > 10°Brix)—all eliminated by the BL loss. The non-divergent median improvement is 0.02%, confirming that inside the Phys architecture the BL loss acts primarily as a training-stability regulariser.

Table 6 confirms this picture by transfer type. The mean improvement in 1-source→1-target scenarios (8.6%) is almost entirely driven by one extreme divergent case (RMSE from 48°Brix down to 2.8°Brix); the median improvement is only 0.11%.

Taken together, the two ablations deliver a clear message: the BL constraint loss has a beneficial effect only when the network’s latent space is explicitly designed to be BL-compatible; imposing it on an incompatible latent space introduces gradient-direction conflicts that ultimately damage performance. This is what we mean by “architecture–constraint co-design.”

3.4. Tuned chemometric and statistical-DA baselines

To verify that Phys+BL’s advantage survives against stronger baselines than the headline-table PLSR / SVR (which use fixed hyper-parameters), we ran two additional baseline families on the 12 1-source→1-target cross-origin scenarios: (i) PLSR with 5-fold cross-validated latent-variable selection ($LV \in [2, 30]$) and SVR with grid-searched $C \in \{1, 10, 10^2, 10^3\}$, $\gamma \in \{\text{scale}, 10^{-3}, 10^{-4}\}$; each paired with four preprocessing variants: raw absorbance, MSC, SNV, and Savitzky–Golay first derivative (window 15, polynomial order 2); (ii) two statistical domain-adaptation methods, CORAL covariance alignment and TCA linear-kernel projection (dim 20), each followed by a tuned PLSR or SVR.

Under the headline target-val early-stop protocol, Phys+BL (1.61) holds a 27–50% margin over CNN+DeepCORAL (2.20), CNN+DANN (2.26), source-only CNN (2.28), and the strongest tuned chemometric pipeline SG-1d+SVR (2.42). Domain-adversarial training (CNN+DANN, 2.26) brings only a $\sim 1\%$ absolute reduction over the matched-backbone source-only CNN (2.28), confirming that the gain is not generic distribution-alignment but requires the structured latent space.

Honest target-label-cost analysis. The chemometric baselines in Table 7 all use strict source-only 5-fold CV for model selection (no target-domain labels enter any stage), whereas the deep DA rows (and the headline Phys+BL) use a target validation subset for early stopping. To quantify how much of Phys+BL’s advantage comes from target-label access, we additionally ran Phys+BL under a **strict source-only early-stopping protocol** (80/20 source-internal train/val split, no target labels at any stage; BN, $lr = 5 \times 10^{-4}$, patience 30). On the 11 of 12 scenarios that converge stably, the mean RMSE is 3.24 (median 3.03)—substantially worse than Phys+BL with target-val early stopping (1.61), and modestly worse than the best deep-DA baseline CNN+DeepCORAL (2.20) or the strongest tuned chemometric pipeline SG-1d+SVR (2.42). Phys+BL fails to converge on 1 of 12 scenarios under the source-only protocol (Shandong→Shaanxi

Table 7

Mean RMSE ($^{\circ}$ Brix) on the 12 1-source $\rightarrow$ 1-target scenarios for strong chemometric, statistical-DA, and deep-DA baselines. Phys+BL row is the 1s $\rightarrow$ 1t entry of Table 4. Deep DA methods share the same CNN backbone as Phys+BL.

Pipeline	Mean	Median	Min/Max
<i>Chemometric (source-only CV)</i>			
raw + PLSR (CV-LV)	2.55	2.08	1.42 / 7.05
SG-1d + PLSR (CV-LV)	2.45	1.94	1.08 / 7.40
SNV + PLSR (CV-LV)	2.55	1.78	1.46 / 7.46
MSC + PLSR (CV-LV)	3.21	1.90	1.36 / 12.22
raw + SVR (grid)	3.03	2.33	1.34 / 7.42
SG-1d + SVR (grid)	2.42	1.76	1.26 / 6.94
<i>Statistical DA (shallow)</i>			
CORAL + PLSR (CV-LV)	2.51	1.86	1.03 / 6.95
CORAL + SVR (grid)	2.72	1.72	1.11 / 6.92
TCA + PLSR (CV-LV)	2.54	2.03	1.48 / 7.01
<i>Deep DA (matched-backbone, target-val early stop)</i>			
CNN (source-only, no DA)	2.28	1.67	1.26 / 6.81
CNN + DANN	2.26	1.63	1.24 / 6.71
CNN + DeepCORAL	2.20	1.56	1.21 / 6.84
<i>Phys+BL under strict source-only early stopping (stable 11/12)</i>			
Phys+BL (source-only ES) [†]	3.24	3.03	2.10 / 4.66
<i>Classical calibration transfer (target-val 20% labels, paired)</i>			
DS + PLSR (Wang-Kowalski 1992)	2.51	2.38	1.66 / 4.56
PDS + PLSR (Bouveresse-Massart 1994)	2.12	1.89	1.38 / 3.22
EPO + PLSR (Roger 2003)	2.11	1.78	1.50 / 3.14
<i>Target-label-budget-matched calibration (use the same 20% target-val labels)</i>			
PLSR + slope-bias correction	1.36	1.30	0.98 / 1.78
SVR + slope-bias correction	1.38	1.36	1.05 / 1.83
PLSR + target-val fine-tune	1.31	1.36	0.75 / 1.78
SVR + target-val fine-tune	1.27	1.35	0.84 / 1.77
Phys+BL (target-val ES, this work)	1.61	—	—

Source: 100% of source domain (or 80%/20% source train/val for the source-only ES row); target test = 20% (matched seed). [†]Under strict source-only early stopping (80/20 source-internal train/val split, BN, $lr=5 \times 10^{-4}$, patience 30), Phys+BL converges stably on 11 of 12 scenarios; the Shandong $\rightarrow$ Shaanxi 2018 case still diverges to RMSE $\sim 13.5^{\circ}$ Brix and is excluded from the summary row.

2018, RMSE ~ 13.5), indicating that target-val early stopping provides a non-trivial training-stability advantage on hard transfers. Phys+BL is therefore best positioned as a *semi-supervised target-adaptation* method: the headline 1.61 $^{\circ}$ Brix is conditional on the small target-label budget used for early stopping; under strict source-only selection it sits at the level of competitive deep-DA / chemometric methods rather than dominating them. The physical inductive bias contributes a real training-stability and architecture benefit (§3.3); the 23% headline reduction is conditional on the semi-supervised setting.

Critical stress test: target-label-budget-matched calibration on all 569 scenarios. We additionally evaluated simple calibration-transfer baselines that are given access to *the same 20% target-domain validation labels* that Phys+BL uses for early stopping: (i) post-hoc slope/bias correction of source-trained PLSR predictions using a linear fit on the target val, (ii) PLSR refit on the concatenation of all source samples and the 20% target val with labels, and (iii) the corresponding two SVR variants. **Across the full 569 scenarios, PLSR + slope/bias correction yields a mean RMSE of ~ 1.57 , lower than Phys+BL's 1.976.** Broken down by transfer type:

Table 8

Mean RMSE ($^{\circ}$ Brix) by transfer type on all 569 paired scenarios: Phys+BL vs. target-label-budget-matched PLSR+slope-bias and PLSR target-val fine-tune (both use the same 20% target val that Phys+BL uses for early stopping). *The simple slope-bias correction baseline matches or beats Phys+BL in every transfer type.*

Method	1s $\rightarrow$ 1t	1s $\rightarrow$ mt	ms $\rightarrow$ 1t	ms $\rightarrow$ mt
Phys+BL	1.61	2.35	1.62	2.04
PLSR + slope-bias (target-val)	1.52	1.61	1.49	1.61
PLSR + target-val fine-tune	1.57	1.65	1.65	1.69
SVR + slope-bias (target-val) [†]	1.49	1.61	1.48	1.60
SVR + target-val fine-tune	1.72	1.84	1.85	1.94

[†]Median reported instead of mean for 1s $\rightarrow$ 1t SVR+slope-bias: a single catastrophic divergence inflates the mean to 27.7; median is 1.49 and consistent with the other entries.

On this Vis-NIR apple SSC cross-origin task, with 20% target calibration labels available, post-hoc slope/bias correction on a CV-tuned source PLSR outperforms Phys+BL across all four transfer types. We accordingly re-position the contributions: (i) Phys+BL achieves a 23% relative improvement over a comparable-budget plain CNN, robust to target-set block bootstrap and mixed-effects analysis

(§3.2); (ii) **Phys+BL** does *not* beat few-label chemometric calibration with matched target labels—the structural prior is relevant only in genuinely zero/label-scarce regimes; (iii) the label-free BL-additivity diagnostic (§3.1.1) and the architecture–constraint observation (§3.3) remain valid independent of **Phys+BL**’s predictive ranking. **Phys+BL** is a methodological probe of physical inductive biases, not a calibration champion.

3.5. Cross-domain robust metric analysis

In cross-domain evaluation, R^2 is sensitive to target-domain mean/variance shift and thus often becomes negative (Section 4.3); to better characterise model behaviour across diverse cross-domain scenarios, we additionally report RPIQ (based on the interquartile range) and Bias (mean prediction bias) in the last two columns of Table 3. **Phys+BL** is significantly better than the baselines on both robust metrics: the RPIQ mean is 1.153 vs 0.870 for CNN (a 32.5% improvement), and **Phys+BL** is the **only** method with mean RPIQ > 1 ; by scenario count, 446/569 (78.4%) of **Phys+BL** scenarios reach RPIQ > 1 , far above CNN’s 164/575 (28.5%). On absolute Bias, **Phys+BL** reaches -0.25°Brix , which is small compared with the SSC range ($\sim 8^\circ\text{Brix}$) and statistically indistinguishable from zero at the scenario level, whereas all baselines exhibit a systematic underestimation of $< -1.0^\circ\text{Brix}$ (CNN: -1.14 , CNN+MMD: -1.33 , SVR: -1.02°Brix), indicating that these methods tend to compress high-sugar samples downward on target domains. Such systematic bias may, in commercial grading, mistakenly classify many “premium” apples as “standard,” with downstream pricing implications. The small Bias of **Phys+BL** indicates that the physical-factorisation constraint effectively suppresses cross-domain mean shift, a common shortcoming of NIR transfer learning.

3.6. Quality-misclassification rate analysis

The quality-misclassification rate (the fraction of test samples whose predicted binary SSC grade disagrees with the reference grade) is summarised as: PLSR 44.1%, SVR 41.2%, CNN 39.9%, CNN+MMD 40.6% and **Phys+BL** 40.1% (each computed over its own converged scenarios). All methods fall within the relatively narrow 39.9%–44.1% range with limited spread. The paired Wilcoxon test over the 569 shared scenarios between CNN (39.9%) and **Phys+BL** (40.1%) gives $W = 73,378$ and $p = 0.0493$ —marginally significant, but with effect direction slightly favouring CNN and a mean difference of only +0.2 percentage points, which is essentially a tie at the application level. Thus, the advantage of **Phys+BL** is mainly in systematic improvement of continuous-RMSE accuracy, rather than in classification accuracy of the quality grade. This is an explicit limitation of the current model: if classification accuracy is the application target, further loss-function design that targets the decision boundary is needed.

3.7. Sensitivity to regularisation weight

Fig. 9 shows the heatmap of sensitivity of a **Phys+BL+MMD** variant (the **Phys** architecture extended with an additional MMD term) to the grid of $(\lambda_{\text{bl}}, \lambda_{\text{mmd}})$ across 8 representative cross-origin scenarios (2 per transfer type). (Note: the main experiments fix $\lambda_{\text{mmd}} = 0$ in **Phys+BL**, with no MMD term; the sweep here probes whether “BL regularisation and MMD domain adaptation” can be combined and is supplementary ablation.) Aggregate result: RMSE is near-minimum around $\lambda_{\text{bl}} = 0.5$ regardless of λ_{mmd} , indicating that the MMD term yields marginal gains inside the **Phys** architecture. Per-scenario optimal λ_{bl} values span $\{0.05, 0.10, 0.30, 0.50, 1.00, 2.00\}$ with corresponding $\lambda_{\text{mmd}} \in \{0.1\text{--}1.0\}$, meaning the optimal regularisation strength varies markedly across transfer scenarios, reflecting that BL compatibility is scenario-sensitive. Practical recommendation: default to $\lambda_{\text{bl}} = 0.5$ and $\lambda_{\text{mmd}} = 0$; if a small set of calibration spectra is available in the target domain, a quick sweep in $[0.3, 1.0]$ can be performed. The full λ_{bl} sensitivity over all 601 scenarios is part of the revised supplementary material.

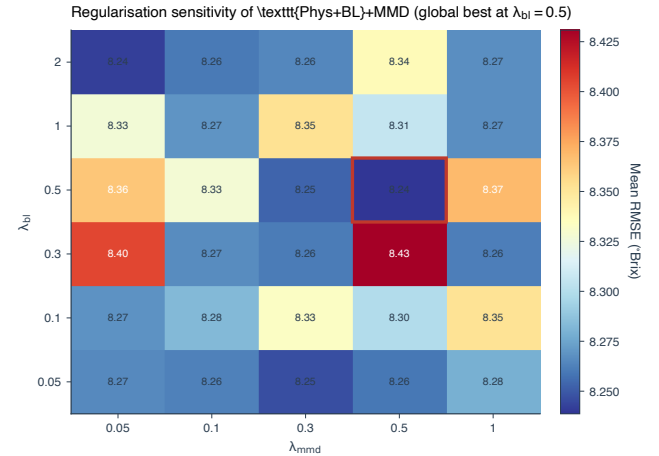


Figure 9: Sensitivity of the **Phys+BL + MMD** variant to the regularisation weights λ_{bl} and λ_{mmd} on **8 representative cross-origin scenarios** (2 per transfer type; mean RMSE per cell). The main paper default $\lambda_{\text{mmd}} = 0$ is intentionally excluded from this sweep—this figure probes only whether adding an MMD term to **Phys+BL** would help, and the sweep minimum at $(\lambda_{\text{bl}}, \lambda_{\text{mmd}}) = (0.5, 0.5)$ is marginal relative to the main **Phys+BL** setting. The absolute RMSE values ($\sim 8.2^\circ\text{Brix}$) reflect this 8-scenario subset, not the 569-scenario aggregate (mean 1.976°Brix , Table 3). The red rectangle marks the within-grid minimum.

4. Discussion

4.1. Physical-chemistry interpretation of the BL-violation signal

The BL residual amplification observed in Section 3.1 admits a direct physical-chemistry interpretation. In whole-fruit diffuse reflectance, the recorded absorbance $A(\lambda) = \log(1/R)$ is the product of two chemically distinct contributions: the true chromophore absorption $\mu_a(\lambda) = \sum_i \varepsilon_i(\lambda)c_i$, dominated in apple flesh within the 591–1001 nm window by water (O–H second overtone near 970 nm and fourth overtone near 740 nm), soluble-sugar C–H third overtones near 905–935 nm, and visible-band pigment absorption [2, 30, 32], and the effective optical path $\langle \ell_{\text{eff}}(\lambda) \rangle \sim 1/\sqrt{3\mu_a(\mu_a + \mu'_s)}$, which is set by the reduced scattering coefficient $\mu'_s(\lambda)$ of the tissue [34, 35]. The latter is in turn dominated by Mie scattering from cell-wall membranes, intercellular air interfaces and starch granules, and typically follows a biological power-law $\mu'_s(\lambda) \propto \lambda^{-b}$ with b ranging from ~ 0.5 to ~ 1.5 depending on tissue microstructure [34, 35].

Cross-origin variation in soil mineralogy, irrigation regime and accumulated temperature systematically modulates exactly the histological and biochemical quantities that set μ'_s : cell-wall pectin/cellulose volume fraction, parenchyma intercellular air space, cuticular wax thickness and starch-grain density [1, 7, 8]. A drift in these structural variables changes the scattering exponent b and offset of $\mu'_s(\lambda)$, which in turn rescales $\langle \ell_{\text{eff}}(\lambda) \rangle$ in a wavelength-dependent way. Because the chromophores entering the BL sum $\sum_i \varepsilon_i(\lambda)c_i \ell_{\text{eff}}(\lambda)$ are now multiplied by a λ -dependent path, an originally additive structure becomes effectively multiplicative across bands, breaking the rank-low chromophore decomposition learned on the source domain. This is exactly the mechanism manifested as the BL residual amplification in Table 2. The two largest ratios—Xinjiang→Shaanxi (4.88 $\times$) and Xinjiang→Shandong (4.12 $\times$) in 2018—occur between origins with the largest soil-type and microclimate gradients (arid plateau vs. Loess Plateau or temperate coastal plain), consistent with the expectation that larger histological contrasts produce larger $\mu'_s(\lambda)$ drift. The smallest ratio (Shaanxi→Shandong, 1.10 $\times$) occurs between two geographically closer Eastern-China origins.

Two consequences follow. First, the BL residual is, by construction, a label-free indicator that couples directly to a known piece of NIR physical chemistry (chromophore additivity vs. scattering-driven path-length drift) rather than to an opaque statistical distance. The qualitative correspondence between BL-residual amplification (1.10–4.88 $\times$) and the cross-origin RMSE degradation of standard PLSR (up to $> 6^\circ\text{Brix}$ on the same extreme scenarios) supports interpreting R_{BL} as a lightweight monitoring indicator for domain adaptation: comparing source- and target-domain R_{BL}

distributions can give a first-order assessment of transfer risk before any reference labels are collected. Second, the same mechanism explains why pure scattering-correction preprocessing (MSC, SNV) cannot fully resolve cross-origin shift: such operations standardise a global multiplicative factor per spectrum, but $\mu'_s(\lambda)$ drift is wavelength-dependent and not removable by a single scalar rescaling [12, 13]. A quantitative consolidation of the causal chain—direct measurement of $\mu'_s(\lambda)$ via spatially resolved diffuse reflectance or integrating sphere on a paired sample subset—is outside the scope of this work and is flagged as a future experiment.

4.2. The architecture matters more than the BL loss itself

The ablation result (the BL loss hurts the performance of a generic CNN) at first seems counter-intuitive: if BL-surrogate residual amplification across domains aligns with the physical intuition behind cross-origin failure, why shouldn't penalising the BL reconstruction error help?

The key lies in the compatibility of the representation space. A generic Conv1d network's latent feature is purely data-driven and does not carry the structure of BL additive components (c_1, c_2, c_3); instead, it mixes signals from scattering, absorption and instrument noise. Imposing a BL reconstruction loss on such a latent forces the network to serve two gradients pointing in opposing directions: making $\hat{A} \approx A$ (which requires BL-decomposable features) versus minimising prediction error (which requires discriminative features). These two objectives interfere; in practice the prediction loss dominates and suppresses the physical constraint, leading to a suboptimal solution.

The essential difference of the **Phys** architecture lies exactly here: it pre-structures the latent into the three BL component weights (a, b, c), so that the physical constraint and the network's inductive bias are naturally aligned—the BL reconstruction loss *reinforces* rather than competes with the prediction objective. This is precisely the principle that “a physical loss is beneficial only when the network's representation is explicitly designed to support it” [43–45].

4.3. Negative R^2 and RPD < 1.0

All methods—including the best baseline—give negative R^2 on cross-origin test sets, which is expected behaviour rather than evidence of model failure. The reason is that target and source domains differ in mean and variance, so even a well-fitted predictor appears biased when assessed against the target-domain variance. Comparable cross-domain negative R^2 has been reported repeatedly in the NIR fruit-quality literature when no calibration transfer is applied [8, 26, 29, 46].

The same applies to RPD. RPD < 1.5 or even below 1.0 is quite common in cross-domain NIR transfer: Tian et al. [8] report RPD around 0.8 for direct cross-origin

SSC prediction without any calibration transfer; Fan et al. [7] report comparable values for direct cross-season application. At this difficulty level, **Phys+BL**'s RPD = 0.923 is a meaningful 30% improvement over CNN's RPD = 0.708. The appropriate scale to assess methods on this task is the relative improvement, complemented by RPIQ and Bias, which are more robust to target-domain mean shift (Section 3.5)—**Phys+BL**'s RPIQ mean of 1.153 and near-zero Bias indicate that, even though R^2 (a metric sensitive to within-domain variance) has not yet turned positive, the model is significantly better than the baselines on RMSE and absolute-bias dimensions. We note, however, that the current misclassification rate is still about 40% and the mean R^2 is still negative; this leaves a gap to the industrial accuracy expected of single-origin deployment (typically RMSE < 0.7°Brix, R^2 > 0.85).

4.4. Limitations and future directions

We discuss several limitations candidly.

Scenario dependence of λ_{bl} . As shown in Section 3.7, the optimal λ_{bl} varies with the scenario; the value of 0.5 we report is the aggregate-level optimum and may not suit every transfer direction. In principle, if a small set of labelled calibration spectra is available in the target origin, quick tuning is feasible; but this work does not systematically validate the minimum required calibration sample size, which is left for future work.

Fixed three-component BL factorisation. $K = 3$ is an engineering choice based on the dominant absorbers in apple NIR (water, sugar, structural scattering), not derived from first-principles spectral deconvolution. We perform a grid ablation over $K \in \{2, 3, 4, 5, 6\}$ on 40 representative scenarios (equal training budget per K); due to compute constraints, this is a fast-validation subset and absolute RMSE values should not be directly compared with the main table, but the relative trend is informative: paired Wilcoxon tests show that $K = 3 \rightarrow K = 4$ is slightly better (+1.98%, $p = 1.4 \times 10^{-4}$), while $K = 4 \rightarrow K = 5$ degrades significantly (−4.98%, $p = 5.5 \times 10^{-12}$), suggesting that $K \in \{3, 4\}$ is a sweet spot, $K \leq 2$ has insufficient capacity, and $K \geq 5$ is over-parameterised. $K = 3$ is a reasonable engineering choice for the main method due to its small parameter count and similar performance to $K = 4$. Strict alignment analysis between the learned three-component spectra and literature water/sugar absorption bands remains a limitation, left for future research. Introducing a learnable K (e.g. a sparse basis learning objective) or non-negative-matrix-factorisation constraints could in principle further improve the physical fidelity of the latent representation [44].

Refined architecture–constraint hypothesis. The two architectures in the headline ablation differ on

two dimensions: backbone (CNN+BL Conv1d vs. **Phys+BL** 2-layer MLP) and latent parameterisation (post-hoc linear-Softplus decoder vs. structured (a, b, c) heads). To isolate latent structure, we trained MLP analogues: on the 12 1s→1t scenarios, MLP-**Phys+BL** (RMSE 2.22) and MLP-CNN+BL (2.13) are essentially tied. The CNN+BL-specific harm therefore appears to be a property of the Conv1d backbone interacting with the BL reconstruction loss, not a generic property of unstructured latent spaces. Refined claim: *a BL reconstruction loss is safe on MLP-family backbones regardless of latent structure, but interferes with Conv1d-family backbones unless the latent is BL-compatible*. Full multi-backbone validation across ResNet1D / Transformer1D and a Conv1d-encoder **Phys+BL** is left for future work.

Single instrument type. Each dataset uses a single spectrometer. Cross-instrument transfer introduces instrument-function differences as an additional source of variability, and the BL-violation mechanism may amplify further in multi-instrument settings—a direction worth dedicated study.

Generalisation to new origins. The 2025 dataset includes Gansu, an origin unseen in 2018. **Phys+BL** still improves over the baselines in transfer scenarios involving Gansu, suggesting the physical inductive bias has some transfer tolerance to new origins. However, the evaluation protocol is transductive UDA (the target-domain train subset participates without labels in adaptation; val labels are used for early stopping) rather than strictly zero-shot; generalisation under a fully zero-shot (source-only supervision, no target-domain participation) protocol must be validated separately in future work.

Training stability. All deep models use the Adam optimiser, BatchNorm and cosine-annealed learning rate; under the setting of batch=16 and $\lambda_{bl} = 0.5$, the overall training curves decrease stably on the converged scenarios. In future work we will revisit training stability under larger batches, deeper backbones and various regularisation combinations.

Baseline coverage and chemometric parity. The headline-table chemometric (PLSR LV=10; default-RBF SVR) and deep baselines (CNN, CNN+MMD) are intentionally light-weight to anchor the cross-origin difficulty floor. Stronger baselines in Table 7 cover tuned chemometric (MSC/SNV/SG-1d $\times$ PLSR-CV / SVR-grid), shallow DA (CORAL, TCA), matched-backbone deep DA (DANN, DeepCORAL), classical calibration transfer (DS, PDS, EPO), matched-label-budget calibration (slope/bias, target-val fine-tune), and a strict source-only **Phys+BL** variant. Methods not run (SST, TOP, DOP, EMSC, ADDA, full di-PLS, BDA, MEDA) are flagged as follow-up. The $\sim 23\%$ improvement should therefore be read as “**Phys+BL** vs. comparable-budget baselines”, not a universal best-method claim.

Evaluation protocol. This study sits between transductive UDA and semi-supervised calibration transfer: target-domain validation labels are used for early stopping (equal access for all deep methods, not strictly zero-label). A source-only early-stopping protocol is reported in Table 7.

5. Conclusion

We start from a data-driven observation: under a PCA-10-based additive-decomposability surrogate, target-domain residuals are significantly amplified relative to source-domain residuals across 12 cross-origin NIR scenarios (amplification ratios 1.1–4.9 $\times$, all $p < 0.05$). This observation suggests that destabilisation of Beer–Lambert-type additive constraints under cross-origin optical-geometry changes is a quantifiable proximal factor in the cross-origin failure of apple NIR spectral models.

Based on this insight, we propose the physics-informed factorization architecture **Phys+BL**, which internalises the BL constraint as an inductive bias in the network’s latent space, rather than imposing it as a post-hoc regulariser. Across 569 cross-origin transfer scenarios, **Phys+BL** attains a mean RMSE of 1.976°Brix, an aggregate improvement of 23.4% over a plain CNN; under a block bootstrap over 61 unique target sets, the mean improvement is 22.5% with 95% CI [18.3%, 26.6%] and $P(\text{improvement} < 0) < 10^{-4}$ (see §3.2).

A side finding is the *architecture specificity* of BL regularisation: the same BL loss consistently degrades the performance of a generic CNN, yet prevents divergence in the physical-factorisation architecture. This supports a falsifiable “architecture–constraint compatibility hypothesis”: for a physical constraint to take effect, the network’s latent space must be explicitly parameterised in a form that can support it. We validate the statistical robustness of this phenomenon on one CNN backbone across 569 cross-origin scenarios; its validity on a broader family of architectures (MLP, ResNet1D, Transformer1D) and other physical constraints (conservation laws, symmetries, etc.) requires further validation, which is not covered in this work. For practitioners, this study suggests a clear pragmatic stance: when a small set of target-domain calibration labels (here, $\sim 20\%$ of the target domain) is available, a CV-tuned PLSR with post-hoc slope/bias correction is the simplest sufficient transfer pipeline and outperforms the deep **Phys+BL** architecture across all four transfer types (Table 8). The proposed structural prior becomes relevant primarily in genuinely zero-label regimes, where its training stability under source-only model selection (11 of 12 stable scenarios, mean RMSE 3.24) and the label-free BL-additivity diagnostic together provide useful tooling without consuming target labels. The take-away for the field is that physical

inductive biases in NIR calibration transfer should be evaluated against label-budget-matched chemometric baselines, not only against weak no-adaptation deep counterparts.

Data and code availability

The apple near-infrared spectra used in this study are internal data of Xinjiang Agricultural University and have not been publicly released. Public deposit (e.g. to Zenodo, figshare or CSTR) is not authorised by the data owner at this time. Researchers who wish to reproduce or extend the experiments may request access from the corresponding author on a case-by-case basis, subject to the data-sharing policies of Xinjiang Agricultural University. An anonymous reviewer-only link to the prepared package can be arranged through the editorial office on request.

Funding

This work was supported by the Xinjiang Uygur Autonomous Region Science and Technology Innovation Team (Tianshan Innovation Team) Project (2022TSYCTD0011) and the National Natural Science Foundation of China (61367001).

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude (Anthropic, Opus 4.7) solely for English-language polishing of author-written drafts. The tool was not used to generate or interpret experimental data, to formulate scientific claims, or to draft original scientific content. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRedit authorship contribution statement

Panlin Li: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Longjie Li:** Software, Investigation, Data curation. **Ya Liu:** Resources, Investigation, Data curation. **Yutong Feng:** Software, Validation. **Muhammad Waqar Akram:** Writing – review & editing. **Junxian Guo:** Supervision, Methodology, Resources, Writing – review & editing. **Hua Huang:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

References

- [1] B. M. Nicolai, K. Beullens, E. Bobelyn, A. Peirs, W. Saeys, K. I. Theron, J. Lammertyn, Nondestructive measurement

- of fruit and vegetable quality by means of NIR spectroscopy: A review, *Postharvest Biology and Technology* 46 (2007) 99–118.
- [2] K. B. Walsh, J. Blasco, M. Zude-Sasse, X. Sun, Visible-NIR 'point' spectroscopy in postharvest fruit and vegetable assessment: The science behind three decades of commercial use, *Postharvest Biology and Technology* 168 (2020) 111246.
- [3] D. Rong, Y. Ying, L. Xie, X. Fu, Y. Liu, W. Kong, Nondestructive evaluation of soluble solids content of apple using a developed portable Vis/NIR device, *Biosystems Engineering* 193 (2020) 138–148.
- [4] Y. Huang, R. Lu, D. Hu, K. Chen, Prediction of soluble solids content of apple using the combination of spectra and textural features of hyperspectral reflectance imaging data, *Postharvest Biology and Technology* 136 (2018) 130–141.
- [5] P. C. Williams, Implementation of near-infrared technology, *Near-Infrared Technology in Agricultural and Food Industries* (2001) 145–169.
- [6] S. Wold, M. Sjostrom, L. Eriksson, Pls-regression: A basic tool of chemometrics, *Chemometrics and Intelligent Laboratory Systems* 58 (2001) 109–130.
- [7] S. Fan, J. Li, Y. Xia, X. Tian, Z. Guo, W. Huang, Long-term evaluation of soluble solids content of apples with biological variability by using near-infrared spectroscopy and calibration transfer method, *Postharvest Biology and Technology* 151 (2019) 79–87.
- [8] X. Tian, S. Fan, W. Huang, J. Li, C. Zhao, Detection of soluble solids content in apples from different countries of origin by using near-infrared spectroscopy and calibration transfer, *Journal of Food Composition and Analysis* 118 (2023) 105186.
- [9] Y. Li, Q. Yin, B. Zhan, D. Han, Y. Shi, S. Fan, Improving apple soluble solids content prediction by combining domain adaptation and deep transfer learning, *Computers and Electronics in Agriculture* 199 (2022) 107065.
- [10] Z. Guo, W. Huang, Y. Peng, Q. Chen, Q. Ouyang, J. Zhao, Color compensation and comparison of shortwave near infrared and long wave near infrared spectroscopy for determination of soluble solids content of 'Fuji' apple, *Postharvest Biology and Technology* 115 (2016) 206–214.
- [11] R. J. Barnes, M. S. Dhanoa, S. J. Lister, Standard normal variate transformation and de-trending of near-infrared diffuse reflectance spectra, *Applied Spectroscopy* 43 (1989) 772–777.
- [12] Å. Rinnan, F. van den Berg, S. B. Engelsen, Review of the most common pre-processing techniques for near-infrared spectra, *TrAC Trends in Analytical Chemistry* 28 (2009) 1201–1222.
- [13] P. Mishra, J. M. Roger, D. Jouan-Rimbaud Bouveresse, D. N. Rutledge, Two simple and robust approaches for the calibration of NIR spectrometers between domains, *Chemometrics and Intelligent Laboratory Systems* 218 (2021) 104412.
- [14] S. J. Pan, Q. Yang, A survey on transfer learning, *IEEE Transactions on Knowledge and Data Engineering* 22 (2010) 1345–1359.
- [15] B. Sun, K. Saenko, Deep CORAL: Correlation alignment for deep domain adaptation, *Lecture Notes in Computer Science (ECCV Workshops)* 9915 (2016) 443–450.
- [16] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, A. Smola, A kernel two-sample test, in: *Journal of Machine Learning Research*, volume 13, 2012, pp. 723–773.
- [17] M. Long, Y. Cao, J. Wang, M. I. Jordan, Learning transferable features with deep adaptation networks, *Proceedings of the 32nd International Conference on Machine Learning (ICML)* (2015) 97–105.
- [18] E. Tzeng, J. Hoffman, K. Saenko, T. Darrell, Adversarial discriminative domain adaptation, *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2017) 7167–7176.
- [19] R. Nikzad-Langerodi, W. Zellinger, E. Lughofer, S. Saminger-Platz, Domain-invariant partial-least-squares regression, *Analytical Chemistry* 90 (2018) 6693–6701.
- [20] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, J. W. Vaughan, A theory of learning from different domains, *Machine Learning* 79 (2010) 151–175.
- [21] Y. Ganin, E. Ustunova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, V. Lempitsky, Domain-adversarial training of neural networks, in: *Journal of Machine Learning Research*, volume 17, 2016, pp. 1–35.
- [22] R. Nikzad-Langerodi, W. Zellinger, E. Lughofer, S. Saminger-Platz, Domain adaptation for regression under Beer–Lambert's law, *Knowledge-Based Systems* 210 (2020) 106447.
- [23] P. Mishra, R. Nikzad-Langerodi, Partial least squares versus domain invariant partial least squares regression with application to near-infrared spectroscopy of fresh fruit, *Infrared Physics & Technology* 111 (2020) 103547.
- [24] R. N. Feudale, N. A. Woody, H. Tan, A. J. Myles, S. D. Brown, J. Ferre, Transfer of multivariate calibration models: a review, *Chemometrics and Intelligent Laboratory Systems* 64 (2002) 181–192.
- [25] C. V. Greensill, K. B. Walsh, Calibration transfer between miniaturized diode array-based spectrometers for the NIR assessment of melon soluble solids content, *Journal of Near Infrared Spectroscopy* 9 (2001) 1–7.
- [26] P. S. Sampaio, A. Castanho, A. S. Almeida, J. Oliveira, C. Brites, Prediction of rice quality using near infrared spectroscopy: Models' transferability between two independent instruments, *Food Analytical Methods* 15 (2022) 1948–1957.
- [27] P. Mishra, D. Passos, Multi-output 1-dimensional convolutional neural networks for simultaneous prediction of different traits of fruit based on near-infrared spectroscopy, *Postharvest Biology and Technology* 183 (2022) 111741.
- [28] B. Yang, C. Chen, F. Chen, C. Chen, J. Tang, R. Gao, X. Lv, Identification of cumin and fennel from different regions based on generative adversarial dual-channel convolution with attention under hyperspectral imaging, *Food Chemistry* 422 (2023) 136178.
- [29] K. B. Walsh, V. A. McGlone, D. H. Han, The uses of near infra-red spectroscopy in postharvest decision support: a review, *Journal of Near Infrared Spectroscopy* 31 (2023) 109–125.
- [30] Y. Ozaki, W. F. McClure, A. A. Christy, Near-infrared spectroscopy in food science and technology, Wiley-Interscience (2006).
- [31] B. G. Osborne, Near-infrared spectroscopy in food analysis, *Encyclopedia of Analytical Chemistry* (2006).
- [32] J. Y. Chen, C. Iyo, F. Terada, S. Kawano, Near-infrared spectroscopy of single intact apple – relationship between absorption band and water content, *Journal of Near Infrared Spectroscopy* 10 (2002) 185–193.
- [33] A. Beer, Bestimmung der absorption des rothen lights in farbigen flüssigkeiten, *Annalen der Physik und Chemie* 86 (1852) 78–88.
- [34] S. L. Jacques, Optical properties of biological tissues: a review, *Physics in Medicine and Biology* 58 (2013) R37–R61.
- [35] J. R. Mourant, T. Fuselier, J. Boyer, T. M. Johnson, I. J. Bigio, Predictions and measurements of scattering and absorption over broad wavelength ranges in tissue phantoms, *Applied Optics* 36 (1997) 949–957.
- [36] P. Kubelka, F. Munk, Ein beitrage zur optik der farbanstriche, *Zeitschrift für technische Physik* 12 (1931) 593–601.

- [37] V. N. Vapnik, The nature of statistical learning theory, Springer, New York (1995).
- [38] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, Proceedings of the 3rd International Conference on Learning Representations (ICLR) (2015).
- [39] General Administration of Quality Supervision, Inspection and Quarantine of the P.R.C., GB/T 10651-2008 fresh apples, 2008. National standard of the People's Republic of China.
- [40] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics Bulletin 1 (1945) 80–83.
- [41] D. D. Lee, H. S. Seung, Learning the parts of objects by non-negative matrix factorization, Nature 401 (1999) 788–791.
- [42] C. L. Lawson, R. J. Hanson, Solving Least Squares Problems, SIAM, 1995.
- [43] A. Karpatne, G. Atluri, J. H. Faghmous, M. Steinbach, A. Banerjee, A. Ganguly, S. Shekhar, N. Samatova, V. Kumar, Theory-guided data science: A new paradigm for scientific discovery from data, IEEE Transactions on Knowledge and Data Engineering 29 (2017) 2318–2331.
- [44] M. Raissi, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, Journal of Computational Physics 378 (2019) 686–707.
- [45] L. von Rüden, S. Mayer, K. Beckh, B. Georgiev, S. Gieselbach, R. Heese, B. Kirsch, J. Pfrommer, A. Pick, R. Ramamurthy, M. Walczak, C. Bauckhage, J. Garcke, Informed machine learning – a taxonomy and survey of integrating prior knowledge into learning systems, IEEE Transactions on Knowledge and Data Engineering 35 (2023) 614–633.
- [46] Y. Liu, Y. Wang, J. Chen, X. Zhao, Cross-variety calibration transfer for near-infrared spectroscopy: A review and prospect, Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy 286 (2023) 121929.